



v0.1.0

SMART HOSPITALITY MANAGEMENT SYSTEM

AI-assisted Windows desktop PMS for hotels, boutique hotels, guest houses, apart-hotels and resorts

Sell Right. Measure Right. Protect the Data.

Windows desktop (.exe)

100% local AI option

Encrypted identity data

Turkish / English

Version 0.1.0 · August 2026 · MIT license · Legal compliance remains the operator's responsibility

12
34

The Program in Numbers

60

Database tables · 56 ORM models ·
versioned with Alembic

1075

Collected tests — all run without network
access; that rule is itself tested

77.6%

Branch coverage — taken from ``pytest --
cov=app --cov-branch``

72

Permissions · 7 default roles · checked in
two separate layers

13

Critical scenarios — overlap, no-show,
quota, malformed JSON

5

AI providers: LM Studio · OpenAI · NVIDIA ·
Anthropic · Mock

14

Application screens in this deck — all
captured on fictional demo data

0

High-severity bandit findings · medium: 0 ·
low: 9

None of these numbers was typed in: ``sunum/olcum.py`` measures them from the source as the deck is built (19 August 2026). What cannot be measured prints as "not measured", never as a remembered figure.



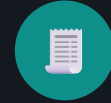
Who It Is For — and What It Is Not

The program runs on a single property's own computer: it is not a cloud subscription, not a channel manager, not an accounting suite.



Target Property

Hotels, boutique hotels, guest houses, aparthotels and resorts with 5–150 rooms. Runs on one machine or a small local network; an internet connection is not required.



What It Covers

The complete front-office cycle from booking to departure: sales, stay, folio, payment, housekeeping, maintenance and management reports.



What It Is Not

Not a channel manager, not a POS, not accounting software. Turkish e-invoice and police-notification integrations are NOT DONE: the data model exists, the connection does not.



Where Responsibility Sits

The software makes data processing easier; compliance with data protection, e-invoicing and guest-notification obligations remains entirely with the operating business.



A Day in Four Screens

The day starts on the dashboard, runs through the front desk and the housekeeping and maintenance teams, and ends with night audit.



Dashboard

Occupancy, ADR, RevPAR and ALOS tiles; critical alerts (overdue departures, open faults, low stock) and a 14-day occupancy curve, all on one screen.



Reservations

Calendar, group bookings, waiting list, channel-tagged records. Night-by-night pricing and overlap prevention run at save time.



Front Desk

Check-in, room card, companions, folio, extra charges, payment and check-out. Early check-in and late check-out are charged in slots.



Housekeeping & Maintenance

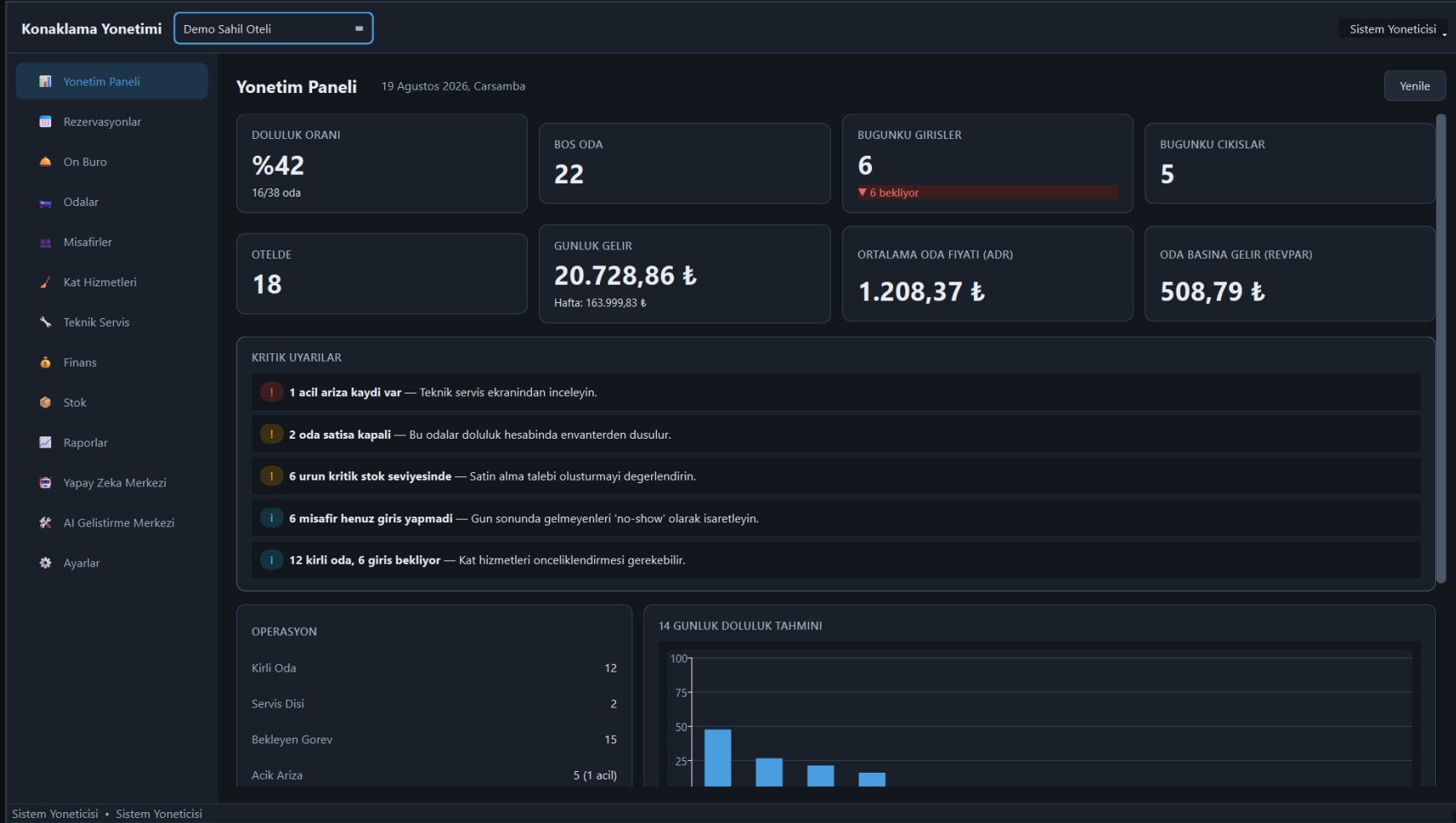
Daily cleaning tasks are generated and assigned; a fault record can take a room out of sale — a blocked room cannot be booked.

Shortcuts: Ctrl+1 Dashboard · Ctrl+2 Reservations · Ctrl+3 Front Desk · Ctrl+4 Rooms · Ctrl+5 Guests · Ctrl+R Reports



On Screen: The Dashboard

Occupancy, vacant rooms, daily revenue, ADR and RevPAR tiles; critical alerts and a 14-day occupancy forecast — the day starts on a single screen.



This and the following screenshots are taken from the program itself; all of the data is synthetic demo content with no real personal data.



On Screen: The Front Desk

Pending arrivals and departures, in-house guests and the uncollected folio balance on one screen; check-in starts from the list with a single button.

Konaklama Yonetimi

Demo Sahil Oteli

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Yonetim Paneli

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On Buro

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On Buro

19 Agustos 2026, Carsamba

Yenile

BEKLEYEN GIRIS

6

Toplam 6 giris

BEKLEYEN CIKIS

5

5 acik hesap

OTELDE

12

6 misafir bekleniyor

ACIK BAKIYE

80.545,36 ₺

Tahsil edilmemis folyo bakiyesi

Bugunku Girisler

Bugunku Cikislar

Otelde

Onay no, misafir veya oda ara

Satir secilmedi

Giris Yap

BUGUNKU GIRISLER

Onay No	Misafir	Oda Tipi	Oda	Gece	Kisi	Durum
DM000018	Melis Fenerli	Standart Oda	108	5	3	Bekliyor
DM000017	Volkan Dagliyol	Deluxe Oda	106	6	3	Bekliyor
DM000016	Berk Bulutlu	Deluxe Oda	206	2	2	Bekliyor
DM000015	Firat Ipekli	Standart Oda	201	4	3	Bekliyor
DM000014	Pinar Ovalik	Aile Odasi	B104	3	3	Bekliyor
DM000013	Baris Karanfilli	Standart Oda	102	1	2	Bekliyor

Sistem Yoneticisi

Sistem Yoneticisi



On Screen: Housekeeping and Maintenance

Daily cleaning tasks are generated with one click, assigned and closed with an inspection step; fault records are tracked with priority and room links.

Housekeeping

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19.08.2026

19 Agustos 2026, Carsamba

Gunun Gorevlerini Olustur

Yenile

BEKLEYEN

10

DEVAM EDEN

5

TAMAMLANAN

5

KONTROL EDILEN

5

GOREVLER

Oda No	Tur	Oncelik	Atanan	Durum	Sure	Kontrolde Goren
103	Gunkuk Temizlik	Normal	Nazli Bulutlu	Devam Ediyor	~30 dk	-
104	Gunkuk Temizlik	Normal	Atanmadi	Bekliyor	~30 dk	-
107	Cikis Temizligi	Yuksek	Nazli Bulutlu	Tamamlandi	20 dk	-
108	Cikis Temizligi	Yuksek	Hazal Cicekli	Kontrol Edildi	20 dk	-
202	Cikis Temizligi	Yuksek	Sena Mercanli	Atandi	~30 dk	-
203	Gunkuk Temizlik	Normal	Efe Findikli	Devam Ediyor	~30 dk	-
204	Carsaf Degisimi	Normal	Sena Mercanli	Tamamlandi	50 dk	-
205	Kontrol	Normal	Efe Findikli	Atandi	~30 dk	-
206	Carsaf Degisimi	Normal	Atanmadi	Bekliyor	~30 dk	-
302	Minibar Dokumu	Normal	Sena Mercanli	Kontrol Edildi	20 dk	-
305	Cikis Temizligi	Yuksek	Nazli Bulutlu	Atandi	~30 dk	-
306	Carsaf Degisimi	Normal	Nazli Bulutlu	Kontrol Edildi	20 dk	-
307	Gunkuk Temizlik	Normal	Nazli Bulutlu	Kontrol Edildi	36 dk	-
308	Kontrol	Normal	Efe Findikli	Kontrol Edildi	42 dk	-

Kat gorevleri: Nazli Bulutlu

Basla

Tamamla

Kontrol Et

Maintenance

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Yeni Ariza

Yenile

ACIK KAYIT

5

toplam 10 kayıt

ACIL / KRITIK

1

hemen mudahale

SATISA KAPALI ODA

1

satisa kapali oda var

TOPLAM MALIYET

10.352,75 TL

isçilik + parca

Kayıt no, oda veya baslik ara

Tum oncelikler

Yalnizca acik kayitlar

ARIZA KAYITLARI

No	Oda No	Kategori	Oncelik	Durum	Ariza	Bildirim	Teknisyen
DA00003	B208	Isitma/Sogutma	Acil	Devam Ediyor	Klima sogutmuyor [SATISA KAPALI]	07.08.2026 10:00	Onur Ruzgarli
DA00007	Ottak alan	BT/Ag	Yuksek	Devam Ediyor	Lobi - ana koridor - Wi-Fi kopmesi	08.08.2026 14:00	Onur Ruzgarli
DA00005	105	Elektronik	Yuksek	Acik	Televizyon acilmiyor	13.08.2026 12:00	Atanmadi
DA00001	Ottak alan	Tesisat	Normal	Acik	Lobi - ana koridor - Banyo giderinde tikanma	07.08.2026 08:00	Atanmadi
DA00009	201	Yapisal	Dusuk	Acik	Duvar boyasi kabarmis	15.08.2026 08:00	Atanmadi

5 / 10 kayıt gösteriliyor

Teknisyen: Onur Ruzgarli

Coz

Kayıtlı Kapat



The Same Room Cannot Be Sold Twice

Overlap prevention is not a warning but a two-layer constraint: the application check and the database constraint enforce the same rule independently.



Two-Stage Check

First the application layer queries availability and produces a readable error; at write time the database constraint applies the same rule again. In a race, the second transaction is rejected.



Half-Open Interval

Dates are modelled as [arrival, departure): the departure day is not a night, so the room can be sold again the same day. Most overlap bugs are born exactly at this boundary.



Night-by-Night Pricing

Season rate, weekend uplift, extra person, discount and tax are computed for each night separately; no single average rate is multiplied by the stay length.



State Machine

Draft → confirmed → checked-in → checked-out; cancellation and no-show are separate paths. Invalid transitions are rejected and fee rules depend on the state.

99 domain tests cover this layer alone: availability 24, pricing 23, state machine 18, money and date value objects 34.



On Screen: Reservations

Search with status and date filters; confirmation number, nights, amount and balance in one list. Cancelled and optional bookings are colour-coded.

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Rezervasyonlar

80 rezervasyon gosteriliyor (toplam 80).

Yenile

Yeni Rezervasyon

Onay no, misafir adi veya oda numarasi ara

Durum: Tumu

Tarih: Tumu

REZERVASYON LISTESI

Onay No	Misafir	Oda	Giris	Cikis	Gece	Durum	Tutar	Bakiye
DM000080	Merve Lalezar	B103	18.08.2026	22.08.2026	4	İptal	13.546,25 ₺	13.546,25 ₺
DM000079	Ece Cicekli	106	18.09.2026	22.09.2026	4	İptal	6.661,24 ₺	6.661,24 ₺
DM000078	Pinar Denizcan	B102	10.10.2026	13.10.2026	3	İptal	3.827,25 ₺	3.827,25 ₺
DM000077	Hazal Denizcan	B102	22.08.2026	25.08.2026	3	İptal	3.150,00 ₺	3.150,00 ₺
DM000076	Nehir Yildizli	208	02.09.2026	03.09.2026	1	Opsiyonlu	2.160,00 ₺	2.160,00 ₺
DM000075	Irem Kumsalli	103	23.09.2026	24.09.2026	1	Opsiyonlu	1.340,00 ₺	1.340,00 ₺
DM000074	Cem Sumbullu	B103	03.11.2026	05.11.2026	2	Opsiyonlu	9.105,00 ₺	9.105,00 ₺
DM000073	Efe Zeytinli	203	26.09.2026	30.09.2026	4	Opsiyonlu	9.103,75 ₺	9.103,75 ₺
DM000072	Deniz Gunesli	B107	25.09.2026	30.09.2026	5	Opsiyonlu	25.042,50 ₺	25.042,50 ₺

REZERVASYON AYRINTISI

Ayrinti için bir rezervasyon secin
Listeden bir satıra tıklayın veya çift tıklayın.

Sistem Yoneticisi • Sistem Yoneticisi



On Screen: The Room Map

A building- and floor-based room map: occupancy and cleanliness are colour-coded; the selected room's details and actions sit in the right-hand panel.

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Odalar

40 oda - 16 dolu, 12 kirli, 2 servis disi

Oda Plani

Liste

Bos - Temiz

Bos - Kirli

Dolu

Servis Disi

ANA BINA - 1. KAT

101

Standart Oda

Dolu - Kirli

102

Standart Oda

Dolu - Temiz

103

Standart Oda

Dolu - Temiz

104

Suit

Dolu - Kirli

105

Deluxe Oda

Bos - Temiz

106

Deluxe Oda

Bos - Kontrol Edildi

107

Aile Odasi

Bos - Kirli

108

Standart Oda

Bos - Kirli

ANA BINA - 2. KAT

201

Standart Oda

Bos - Temiz

202

Aile Odasi

Bos - Kirli

203

Deluxe Oda

Dolu - Temiz

204

Deluxe Oda

Bos - Temiz

205

Deluxe Oda

Bos - Temiz

206

Deluxe Oda

Bos - Kontrol Edildi

207

Standart Oda

Dolu - Temiz

208

Aile Odasi

Dolu - Kirli

ANA BINA - 3. KAT

301

Aile Odasi

Bos - Kirli

302

Deluxe Oda

Bos - Kontrol Edildi

303

Standart Oda

Bos - Temiz

304

Deluxe Oda

Dolu - Temiz

305

Deluxe Oda

Bos - Kirli

306

Aile Odasi

Bos - Kontrol Edildi

ODA AYRINTISI

101 numarali oda

ODA TIPI

Standart Oda

BINA / KAT

Ana Bina - 1. Kat

DOLULUK

Dolu

TEMIZLIK

Kirli

MISAFIR

Dilek Sumbullu

CIKIS

20.08.2026

ACIK ARIZA

Acik ariza kaydi yok

OZELLIKLER

Klima, Elektronik Kasa, Ucretsiz Wi-Fi, LED Televizyon, Sac Kurutma Makinesi, Dusakabin, Sehir Manzarasi, Engelli Erisimine Uygun

Temiz yap

Kirli yap

Servis disi yap

Sistem Yoneticisi • Sistem Yoneticisi

Folio — Never Deleted, Only Voided with a Reason

In financial records, correcting is not deleting. The program never removes a charge line; it voids it with a reason and the audit trail survives.



The Trail Survives

A mistaken charge line is voided and its reason recorded. The night audit total and the history therefore stay explainable at any later date.



Payments and Refunds

Cash, card and transfer payments; refunds, deposits and damage records. The balance is derived from the folio; an inconsistent amount is rejected.



Early In / Late Out

Charged in slots: a 4-hour late check-out is two slots and equals 50% of the nightly rate. The rule is pinned down by a test.



Night Audit

Arrivals, departures, in-house guests and cash movements are collected in the night audit report and exported as PDF, Excel or CSV.

There is no separate Finance screen yet: payments and folio work happen on the Front Desk, financial summaries under Reports.



Identity Data Stays Encrypted — and Searchable

You cannot search an encrypted column. The program resolves that dilemma with a blind index: the data stays encrypted, the search runs over a keyed digest.



Field-Level Encryption

National ID and passport numbers are written to the database encrypted. Verified with a raw SQL query: the column shows no readable number.



Blind Index

Search runs on an HMAC-SHA256 digest of the number: exact match works, the value is never exposed. Partial search is deliberately unsupported.



Unmasking Is a Separate Right

Seeing the unmasked number is a separate permission; not every user who can view the record holds it, and each unmasking is written to the audit log.



Guest CRM

Profile, stay history, preferences, VIP and blacklist, consent records. The summary currently counts only the primary guest; companion nights are excluded.

API keys, e-mail addresses, phone, ID and card numbers are masked automatically in logs; the masking is protected by its own test.



On Screen: Guest Profile and Privacy

The ID number is stored encrypted and displayed masked; the 'Show' button writes every reveal to the audit log with the user name and a timestamp.

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Misafirler60 kayıt

MISAFIR LISTESI

Ad, soyad, e-posta veya telefon ile ara

Ad Soyad	Telefon	E-posta	VIP	Konaklama	Son Ziyaret
! Baris Zambakli	+90 5XX XXX XX ...	baris.zambakli044@ornek-...	Standart	1	14.08.2026
! Elif Sabahci	+90 5XX XXX XX ...	elif.sabahci028@ornek-...	Standart	1	06.07.2026
! Hakan Nergisli	+90 5XX XXX XX ...	hakan.nergisli012@ornek-...	Standart	0	-
! KARA LISTE - ORNEK ...	+90 5XX XXX XX ...	ornek.kayit01@ornek-...	Standart	2	01.08.2026
Aylin Karanfilli	+90 5XX XXX XX ...	aylin.karanfilli040@ornek-...	Standart	1	11.07.2026
Aylin Nergisli	+90 5XX XXX XX ...	aylin.nergisli017@ornek-...	Standart	0	-
Baris Karanfilli	+90 5XX XXX XX ...	baris.karanfilli042@ornek-...	Standart	0	-
Berk Bulutlu	+90 5XX XXX XX ...	berk.bulutlu022@ornek-...	Standart	0	-
Burak Denizcan	+90 5XX XXX XX ...	burak.denizcan015@ornek-...	Standart	0	-
Cem Ipekli	+90 5XX XXX XX ...	cem.ipekli001@ornek-...	Standart	1	16.08.2026
Cem Sumbullu	+90 5XX XXX XX ...	cem.sumbullu047@ornek-...	Standart	0	-
Ceren Ruzgarli	+90 5XX XXX XX ...	ceren.ruzgarli043@ornek-...	Standart	1	29.06.2026
Ceren Sabahci	+90 5XX XXX XX ...	ceren.sabahci013@ornek-...	Standart	0	-
Deniz Bulutlu	+90 5XX XXX XX ...	deniz.bulutlu008@ornek-...	Standart	0	-
Deniz Gunesli	+90 5XX XXX XX ...	deniz.gunesli004@ornek-...	Platin	0	-
Dilek Aydinlik	+90 5XX XXX XX ...	dilek.aydinlik026@ornek-...	Standart	1	03.08.2026
Dilek Karanfilli	+90 5XX XXX XX ...	dilek.karanfilli054@ornek-...	Standart	0	-

Sayin Baris Zambakli

UYARI NOTU

GenelKonaklamalarTercih ve NotKVKK

ILETISIM VE PROFIL

E-posta baris.zambakli044@ornek-test.local

Telefon +90 5XX XXX XX XX (D044)

Cep Telefonu +90 5XX XXX XX XX (D044)

Dogum Tarihi 25 Aralik 1995 (30 yasinda)

Uyruk Ingiltere

Dil Tercihi Ingilizce

Adres Ornek Mahallesi, Deneme Sokak No: 7

Sehir / Ulke Bursa / Ingiltere

VIP Seviyesi Standart

Kurumsal Musteri -

Acente -

Konaklama Ozeti 1 konaklama - 5 gece - 10.494,00 ₺

KIMLIK BILGISI

Pasaport BY1***161

KVKK: Kimlik numarası şifreli saklanır. Acik her görüntüleme kullanıcı adı ve zaman damgasıyla denetim gunlugune yazılır.

Sistem Yoneticisi • Sistem Yoneticisi



Measurement — Hotel Metrics

Occupancy, ADR, RevPAR and ALOS come from one query layer: the number on screen and the number in the report are produced by the same code.



KPIs and Alerts

Occupancy, ADR, RevPAR, ALOS, cancellation and no-show rates. Critical alerts sit at the top: overdue departures, open faults, low stock.



14-Day Occupancy

The next two weeks are drawn from confirmed reservations. This is not a demand-forecasting model but a projection of bookings already on the books.



Channel and Room Type

Revenue is split by channel (direct, agency, online) and by room type; what each channel actually brings in is visible in a single table.



PDF · Excel · CSV

Every report exports in three formats. CSV is written with a UTF-8 BOM so that Excel opens accented characters correctly.

106 reporting tests also cover export formats and the empty-data case: with no data the report does not crash, it renders empty.



On Screen: Reports

Occupancy, channel and room-type revenue, night audit, housekeeping and a KPI summary; every report can be exported to PDF, Excel and CSV.

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Ozel aralik

Baslangic

01.08.2026

Bitis

19.08.2026

Raporu Olustur

RAPOR TURU

Doluluk

Gelir (Kanal)

Gelir (Oda Tipi)

Gun Sonu Kapanis

Giris - Cikis

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KPI Ozeti

Gun bazinda toplam, arizali, satilabilir ve dolu oda sayilari.

DOLULUK

%15,4

117 / 759 oda gecesi

ADR

1.495,64 ₺

Yalnizca oda geliri

REVPAR

230,55 ₺

Toplam: 195.285,69 ₺

ALOS

3,08 gece

Ortalama konaklama suresi

IPTAL ORANI

%2,6

Gelmeme: %5,3

RAPOR

Doluluk Raporu (Demo Sahil Oteli - 1 Agustos 2026 - 19 Agustos 2026 (19 gece))

Rapor icinde ara

Tarih	Toplam Oda	Arizali	Satilabilir	Dolu	Bos	Doluluk
01.08.2026	40	0	40	4	36	%10,00
02.08.2026	40	0	40	3	37	%7,50
03.08.2026	40	0	40	3	37	%7,50
04.08.2026	40	0	40	1	39	%2,50
05.08.2026	40	0	40	1	39	%2,50
06.08.2026	40	0	40	0	40	%0,00
07.08.2026	40	0	40	0	40	%0,00
08.08.2026	40	0	40	0	40	%0,00
09.08.2026	40	0	40	1	39	%2,50
10.08.2026	40	0	40	1	39	%2,50
11.08.2026	40	0	40	2	38	%5,00
12.08.2026	40	0	40	1	39	%2,50

19 satir

Disa aktar: PDF Excel CSV Klasoru Ac

Sistem Yoneticisi • Sistem Yoneticisi



The AI Advises — It Never Writes

The AI layer is the security boundary between the database and the model: no personal data leaves, and the model never changes a record on its own.



Prompt Screening

E-mail addresses, phone numbers and long digit strings are masked before the call; if any personal data remains after masking, the request is never sent.



Local Option

With LM Studio the model runs on your own machine: no cloud, no API bill, no data leaving the building. Cloud providers are optional.



Advice Only

Rate suggestions, guest-review sentiment analysis and message drafts are produced. None is applied automatically; the decision stays with the user.



Badge and Cost

Every output is marked with an "AI generated" badge; token count, duration and estimated cost are shown — free on a local model.

If the primary provider fails, fallback happens only on TRANSIENT errors. An invalid API key is permanent: it is reported, not hidden.



On Screen: The AI Center

Ready-made tasks and free-form questions; provider and model are chosen on the same screen. Every answer carries an 'AI-generated' badge.

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Saglayici

Imstudio

Model

Otomatik (gorev turune gore)

Baglanti basarili (Imstudio)

Baglantiyi Test Et

HAZIR GOREVLER

Gunluk Ozet

Doluluk Analizi

Fiyat Onerisi

Mesaj Taslagi

Yorum Analizi

Serbest Soru

Taslak turu

Rezervasyon onayi

Yapay zeka ciktilari hata icerebilir.
Kritik kararlarda dogrulayin.

SOHBET

Soldaki hazır görevlerden birini seçin ya da sorunuzu aşağıya yazın.
Yanıtlar 'AI tarafından oluşturuldu' rozetiyle işaretlenir.

Sorunuzu yazın veya bir görev için metni buraya yapıştırın (Ctrl+Enter: gönder)

Gonder

Sistem Yoneticisi

Sistem Yoneticisi

AI Development Centre — A Restricted Sandbox

The AI can be given coding tasks; what it may do is bounded by a command policy, a sandbox and a quality chain.



Command Policy

Two layers: (1) an allow-list decides which command may run, (2) a target check decides which file it may touch. `.env``, key files and the guest database are closed to every reader, whichever one is used.



Sandbox and Diff

Writes happen only under the sandbox root; every change is shown as a diff first, nothing is written without approval, and lost-update protection is in place.



Git Guard

The task runs on its own branch, a checkpoint is taken before it starts, and failure rolls back automatically. Merging to main needs separate approval.



Quality Chain

Format → lint → types → tests → security. No change is committed until the tests pass, and the user cannot relax that rule.

187 tests cover this module. The layer that actually executes the command has low coverage; that gap is stated openly in the test report, with its measurement.

Security — Measured, Not Assumed

The security review is not a list of claims: each item rests on a test or on the output of a tool that was actually run.



Authentication

Passwords are hashed with Argon2id; brute-force lockout, session timeout and a uniform error message that prevents user enumeration are in place.



Authorization Twice

Permissions are checked both in the UI (menu, buttons) and in the service layer. Hiding a screen is not enough; the unauthorized service call is rejected too.



Secret Handling

API keys live in Windows Credential Manager; they are never written to the database, the source or error messages. The repository was scanned for secrets.



Audit Log

Critical operations are written to an append-only audit log: who did what, when, and to which record.

bandit: 0 high, 0 medium and 9 low-severity findings — each low finding carries a written rationale. The numbers are measured as the deck is built.

The Reviews Found Real Holes

The value of a security review is in what it finds. Everything below was fixed, each has a regression test, and the fixes are described in docs/SECURITY_REVIEW.md section 8.



Missing Permission Check

Taking a room out of sale skipped the permission check: a user without maintenance rights could block a room. Closed; guarded by ``tests/application/test_operations_services.py``.



No Ownership Check

AI chat history did not verify ownership; one user could read another's conversation. Closed; guarded by ``tests/application/test_ai_service.py``.



Policy Bypasses

Bypasses were found in the command policy and closed. The last one came out of an independent pre-release review: an allow-listed reader (``head``, ``findstr``) could read ``.env`` with no approval. The decision now looks at the file touched, not at the command's name.



Financial Inconsistency

Arithmetic errors in folio and payment handling were corrected, and a key error being swallowed during provider fallback was fixed. Money never travels as ``float``, only as ``Decimal`` — and that is tested.

No commit hashes are cited on purpose: the evidence for a fix is the test that guards it. The same document lists what was not fixed — the number generators are not concurrency-safe.



The Numbers Were Run, Not Remembered

1075 tests pass and none of them touches the network — and that fact is itself protected by a test. The figure was measured while this deck was generated.



Layer by Layer

Domain 99 · security 75 · infrastructure 145 · application 158 · AI 123 · UI 182 · reporting 106 · development centre 187.



13 Critical Scenarios

Overlapping booking, cancellation, no-show, early arrival, blocked room, wrong payment, unauthorized access, dead model, quota, invalid key, malformed JSON, empty report — all covered.



Gaps Written Down

The startup path (login screen, main window) and the CLI stay uncovered; backup has low coverage. None of this is hidden — it is listed with its measurements and explained in docs/TEST_REPORT.md.



CI on Every Push

GitHub Actions (.github/workflows/ci.yml): format, lint, types, tests, security scanning, migration consistency and a "did a sensitive file enter the repo" gate. `ruff check app tests` currently reports 0 findings.

The report is a snapshot: 19 August 2026, version 0.1.0, Windows 11, Python 3.11. The numbers are measured at generation time and change when the code does.



Architecture — Dependencies Flow One Way

ui → application → domain ← infrastructure. The business rules know neither the database nor the UI, which is why the 99 domain tests run in seconds.



Framework-Free Domain

The domain layer imports neither the ORM nor the UI toolkit. The rule is not only documented but enforced by an import test.



Timezone-Aware

A custom column type stores every timestamp in UTC. Offset and daylight-saving bugs are solved in one place instead of scattered through the data layer.



Migratable Schema

60 tables are versioned with Alembic and CI verifies schema/migration consistency on every change. Enumerations are stored as text, not as integers.



Page Registry

Adding a screen does not touch the main window: one line is added to the registry. Unfinished screens are declared in the same place.

Setup, Backup and Packaging

One command to install, one to back up, one to package. The database and logs live next to the application, so the installation stays portable.



One-Command Setup

The setup script validates the Python version, creates the virtual environment, writes .env with a random session key, applies migrations and creates the admin account.



Consistent Backup

The SQLite backup uses VACUUM INTO; copying the file in WAL mode produces an inconsistent backup. The current database is kept aside before any restore.



Windows Package

The build script produces an .exe with PyInstaller; on first launch a wizard prepares the database and shows the admin password once.



Maintenance Commands

bootstrap · seed-demo · backup · restore · check-ai · doctor. check-ai tests the provider connection, doctor the health of the installation.

A demo installation ships five role accounts (manager, front desk, housekeeping, maintenance, accounting). In a real deployment demo data and accounts must be removed.



What Is Not Done — Stated Just as Clearly

If a feature is on the roadmap, it means it should not be used in production today. The program does not hide its gaps from the user.



e-Invoice and Notification

NOT DONE. The invoice table has the fields and the settings file the keys; the authority integration, XML generation and notification flow do not exist.



Finance · Stock · Staff

Business logic and reports exist, dedicated screens do not. Opening one shows what is planned and how to do the same job today.



Number Generator

Reservation and folio numbers are produced with `max()+1`; under concurrent use the second write is rejected and must be retried. The fix lands in v0.2.



Cloud Providers

The NVIDIA and Anthropic adapters are written but never exercised against the real API — that needs a key. LM Studio was verified with a real request.

Document question-answering (RAG) and visual document analysis are also unfinished: the data model exists, the workflow is not wired.

Roadmap

The order is deliberate: first the screens that complete daily operations, then depth in AI, and regulatory integrations last.



v0.2 — Operations

Finance, stock and room-type/rate-plan screens; staff and shift management; the concurrency fix for number generation; a group booking interface.



v0.3 — Artificial Intelligence

Completing the document indexing flow, visual document analysis (with a data-protection assessment), demand forecasting and dynamic rate suggestions.



v0.4 — Integrations

The e-invoice integrator connection, guest notification, a channel manager draft and the e-mail/SMS notification layer.



v1.0 — Production

PostgreSQL production setup, multi-property testing, an independent security audit, performance testing with 10,000+ reservations and user acceptance tests.

Most open to contribution: regulatory integrations, workflow feedback from real hotels, and the English interface translation.



RUN THE HOUSE KNOWING WHAT YOU SOLD

60 tables · 1075 tests · 77.6% coverage · encrypted identity data · a local AI option

A program that writes down its gaps too.

User guide

Test report

Security review

Roadmap